

Agile Collaborative Research:

Action Principles for Industry-Academia Collaboration

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// A model for industry-academia collaborative practice research defines 10 factors for successful projects and 10 action principles for collaboration management. //



(see Figure 1).³ Inspired by the work of Donald Schön,⁴ CPR brings acting practitioners and reflecting researchers together to obtain desired improvements by using methods such as action research, experiments, and practice studies.

However, industry and academia have different objectives (new products and sales versus new knowledge and fundraising) that complicate CPR management. Although it's relatively easy to agree on challenges and goals, viewpoints differ regarding variables such as relevance, rigor, time horizons, planning practices, and predictability.

Recent research has characterized these viewpoints⁵ as well as success factors in collaboration management.^{3,6,7} In comparing these findings with our own experiences setting up a research collaboration, we recognized an opportunity to offer guidance on how to manage CPR efforts. To this end, we present a model for successful CPR. The model distills eight years' experience with a CPR effort between a telecommunications company and an academic research institute.

The Ericsson-ITU Collaboration

Ericsson is a global developer of telecommunication equipment and services. Over the past decade, software's importance to Ericsson's business has increased dramatically, which in turn has stimulated the company's collaborations with academic software researchers worldwide.

In 1997, government, industry, and academic institutions in western Sweden joined forces to collocate a new research institute with the emerging high-tech industry cluster in Gothenburg's old wharf area. The institute's mission was to perform "research and education within a small set of cross-

INCREASED COMPETITION AND globalization in the software community are motivating industrial and academic institutions to improve their collaborations. More than ever, both sides perceive traditional methodologies, such as innovation diffusion or value chains that transfer basic research to commercial applications via

research institutes and startups, as insufficient for current industry challenges and research goals.^{1,2} These approaches effectively convey innovations but don't support fundamental process improvements.

Collaborative practice research (CPR) is an approach Lars Mathiassen defined to support these improvements

disciplinary themes of a kind not provided by universities downtown.” Ericsson was a main driver in the subsequent formation of IT University (ITU) in 2000, making an initial investment the next year to engage and align ITU’s research and teaching activities with the company’s goals. The engagement eventually led to the 2006 formation of a collaborative research center for software architecture and quality (SAQC). As SAQC’s prime financial contributor, Ericsson owns all of the intellectual property it creates.

Digging through historical project data and talking with people involved in or affected by the collaboration, we identified four phases in SAQC’s life:

- *initialization* (2001–2004), distinguished by the selected team’s enthusiasm and start-up projects;
- *alignment* (2005–2006), consisting of activities to align the local collaboration with Ericsson’s global strategies;
- *formalization* (2007–2008), which secured Ericsson management’s involvement and refined the collaboration structure; and
- *utilization* (2009–present), in which management is institutionalizing the approach and preparing for growth.

We discuss each phase, including tables that summarize the main CPR projects and key events. Each project or event has a case name and number. The text references them using the form (#*n*).

Initialization

Table 1 lists CPR projects and key events during the initialization phase.

In 2001, Ericsson Mobile Data in Gothenburg signed a formal agreement with ITU to sponsor software engineering research (#1). To this end, Ericsson and ITU recruited several skilled engineers and researchers in strategic areas and encouraged them to collabora-

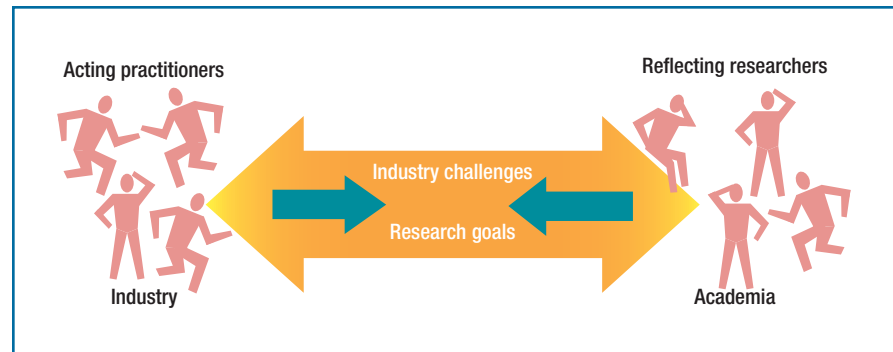


FIGURE 1. Collaborative practice research (CPR). Skilled professionals from industry and academia bring the strengths of each community to solving industry research needs.

rate through informal get-togethers. Researchers presented their expertise, company representatives described their needs, and both parties brainstormed potential collaborations.

Four software quality projects originated during initialization (#2, #5, #7, and #8). The first was the Software Process Improvement (SPI) project (#2), in which an enthusiastic Ericsson SPI manager and an ITU professor recognized for SPI expertise discussed ways to use CPR to improve SPI at Ericsson.⁸

The get-togethers inspired engineers and researchers but didn’t always provide enough structure to sustain CPR (for example, #5). In particular, projects lacked clear direction. Follow-up was ad hoc; strategies, projects, and results weren’t formally communicated; and organizational support was lacking for collaborators to understand or position themselves relative to a broader perspective. That said, managers held regular meetings to decide how to proceed.

Projects #2, #7, and #8 addressed industry needs, but only #2 and #7 matched specific business units’ prioritized needs at that time. The projects addressed needs on both general and unit levels, and they had an impact on deployments that benefited industry practices: #7 successfully applied a new testing tool in practice, and #2 successfully implemented process improvements.

The projects that didn’t address unit-level needs either never took off (#5) or applied only to future CPR projects (#8).

The collaborative teaching environment that emerged during initialization was a positive side effect of the research. Ericsson appreciated the integrated education forms reflected on projects #3 and #4 enough to establish guest lectures (#6) that capitalized on them.

Alignment

Table 2 lists CPR projects and key events during the alignment phase.

After three years of projects, we needed to relate collaboration goals with Ericsson’s global software research strategies. Managers at Ericsson and ITU met to clarify the collaboration’s purposes. They formed SAQC with visions, goals, and focus areas specific to Ericsson needs (#11, #13, and #19). From this point, all collaboration projects had to connect to global research activities and describe benefits to Ericsson. To make this happen, Ericsson management authorized its engineers to spend more time on CPR projects.

The university researchers began to see that “pushing” solutions toward industry wasn’t a viable CPR approach. An influential talk (#9) led to a change in strategy from push to pull. Rather than trying to adapt and implement

TABLE 1

Collaborative practice research (CPR) projects and key events during the initiation phase (2002–2004).

Project/event #	Case data	Type	Date	Explanation	Outcome/Effect
1	Collaboration agreement	Key event	Dec. 2001	Ericsson and ITU sign a framework agreement.	Collaborations started.
2	Software process improvement (SPI) project	CPR project	Jan. 2002	Action research/PhD research. Goal is to improve SPI practices at Ericsson.	Project concluded. Several SPI practices implemented; 15 publications; PhD defense in 2006 (#15).
3	New, integrated university environment	Key event	Sep. 2002	ITU sets up an education environment that integrates teachers with students and research with education.	Industry appreciated environment setup.
4	Innovative teaching setup	Key event	Sep. 2002	ITU focuses on problem-based learning and organizes project classes to be similar to industry work.	Industry appreciated teaching setup.
5	Common criteria project	CPR project	Nov. 2003	Umbrella research. Goal is to align activities of newly recruited researchers toward a framework for IT security evaluation and improvement.	Project interrupted.
6	Guest lecturing SPI courses	Key event	Jan. 2004	Ericsson employees start giving main part of the lectures in the SPI university courses at ITU.	Ericsson employees gave SPI lectures at ITU from 2004 to 2010.
7	Trace analysis project	CPR project	2004	Proof-of-concept research. Goal is to develop a tool to visualize execution traces in an Ericsson product.	Project concluded. New tool used in practice; one publication; new CPR project (#29).
8	Flawless project	CPR project	2004	Proof-of-concept research. Goal is to adapt academically mature, model-checking technology to processes and tools used within Ericsson.	Project concluded. Identified improvement need; two publications.

what researchers knew and liked, they would focus on identifying and addressing needs in close collaboration with the business units.

ITU researchers introduced a new format for the get-togethers. Ericsson engineers spent 10 minutes each presenting prioritized problems. Each researcher and engineer then selected the problem he or she found most interesting and turned it into a project proposal that described the project's value to stakeholders and industry in general.

The changes in strategy and meeting format immediately resulted in 11 proposals, of which nine were accepted for presentation to the steering group and three were eventually approved as research projects (#10, #12, and #16). A CPR-based PhD defense (#15) also inspired collaboration. Project monitoring improved. When disturbances, such as reorganizations, caused engineers or researchers to lose focus on a project, the steering group quickly terminated it (for example, #17 and #18) and initiated a new project in its place. Successful

projects during this period focused on not only industry needs but also industry goals (for example, #20 and #21).

Project direction also improved after SAQC was formed. By the end of this phase, researchers and engineers who had worked within SAQC over a longer time were convinced of CPR's benefits. More projects proved their value (for example, #10, #12, #16, #20, and #21). However, there was still no organized way to communicate research activities or their results, and the decision-making process was ill-defined.

CPR projects and key events during the alignment phase (2005–2006).

#	Case data	Type	Date	Explanation	Outcome/Effect
9	Influential talk from visiting professor	Key event	Feb. 2005	Visiting professor Claes Wohlin (with 20 years' industry-academia collaboration experience) stresses importance of industry-motivated research.	Researchers encouraged to change from push to pull strategy in industry collaborations.
10	QuickCheck project	CPR project	May 2005	Proof-of-concept research. Goal is to use QuickCheck tool to integrate testing activities with development activities.	Project concluded. New tool used in practice; one publication; new CPR project (#12).
11	Influential ideas by visiting professor	Key event	May 2005	In discussions with Anna Sandberg, visiting professor Lars Pareto suggests formalizing the research collaboration into the Centre for Software Quality Research (SAQC).	Starting point for the first steps toward a formalized collaboration called SAQC.
12	Project to generate QuickCheck evaluation tests from interwork descriptions	CPR project	Nov. 2005	Tool evaluation and adaptation research. Goal is to use QuickCheck to automatically generate and execute test cases.	Project concluded. Updated tool used in practice; one publication; new European Union project (#29).
13	Formation of reference group: first group work session	Key event	Dec. 2005	Ericsson employees present their needs in a workshop; researchers engage in group work sessions to generate initial project proposals that address these needs.	Reference group institutionalized with SAQC.
14	Model-driven development course	Key event	Dec. 2005	Ericsson initiates training on modern software development processes: senior line and project managers are to participate in a university course with students.	University course held in 2006, 2007 2008, and 2009; new CPR project (#21).
15	CPR-based PhD defense	Key event	Jan. 2006	Anna Sandberg defends her doctoral thesis "Making SPI Happen," which describes several CPR projects.	Further collaboration between industry and academia inspired, and use of action research increased.
16	Consistency project	CPR project	Feb. 2006	Qualitative case study research. Goal is to improve the consistency between programs and design models.	Project concluded. Architectural understanding; one publication; new CPR project (#27).
17	Coupling and cohesion project	CPR Project	Mar. 2006	Proof-of-concept research. Goal is to visualize architectural qualities from trouble reports with a new academic tool (CCVisu).	Project interrupted.
18	Project to evaluate adapted software performance evaluation (SPE)	CPR project	Sep. 2006	Method evaluation research. Goal is to evaluate whether an SPE method applied to problems experienced by an Ericsson unit.	Project interrupted.
19	Ericsson software research setup	Key event	Aug. 2006	Ericsson-ITU collaboration is formally named SAQC and is connected to Ericsson's global software research activities.	Research aligned with global company strategy.
20	Metrics project, part I	CPR project	Sep. 2006	Empirical, quantitative, theory-building research. Goal is to develop a statistical prediction model for trouble reports based on project data.	Project concluded. Organizational understanding; three publications; new CPR project (#28).
21	Practitioner integrated learning (PIL) project	CPR project	Sep. 2006	Action research. Goal is to train managers in model-driven development, using new PIL format for continuing professional development.	Project concluded. New teaching format; 15 managers trained; one publication.
22	System management training	Key event	Oct. 2006	Researchers give introductory training on requirements elicitation and agile software processes to approximately 50 Ericsson system managers.	Engineers trained in state-of-the-art requirements practices.

Formalization

Table 3 lists CPR projects and key events during formalization. This phase addressed weaknesses in decision and communication structures.

Project #25 introduced reporting procedures. A steering group with representatives from ITU and Ericsson's software divisions and software research functions began meeting quarterly to discuss project proposals

continuity (for example, #27 and #33). Project #28 changed focus from prediction modeling to automation so that it better matched a specific unit's goals, and #33 started up quickly when an Ericsson management team decided to improve agile processes.

The metrics project (#28) now counted two years of continuous research and began showing deployment impacts. Posters and public presenta-

project resulted in a publication, recognized for excellence by both academia (IEEE Computer Society Distinguished Paper Award at the 2009 Model-Driven Engineering Languages and Systems Conference) and industry (Best Ericsson Software Research Paper Award 2009) (#41).⁹

Three hitherto parallel projects (#36, #37, and #39) aligned in a joint research funding proposal to achieve a shared goal (#38). This led to a common view among participants of how SAQC contributed to a larger perspective, which was summarized and visualized as a new SAQC profile (#40). Presentations, posters, progress reports, and proposals were now aligned, which helped connect the research projects to Ericsson's business goals.

Administrative activities had become routine, letting collaborators focus on core activities to understand unit needs, align projects with company goals, improve practice, and increase benefits. These activities, in turn, contributed to the academic body of software engineering knowledge.

Management activities were closely allied with research results. Internal project management was insufficient for ensuring relevant results. Research projects must also continually engage higher management and communicate research results to stay relevant and produce clear value.

Success Factors

Together with five Ericsson managers involved in the project, we reflected on project data and developed 10 factors that we regard as particularly influential to SAQC's success (see Table 5). The first five relate to research activities and the second five, to research results.

We evaluated the average project's maturity and its management in each phase with respect to each factor. Evaluation proceeded by iterative discussions, in which we used a

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and decide which ones to pursue. Get-togethers became quarterly reference-group meetings led a member of Ericsson's global software research board (#13).

Ericsson renewed its SAQC sponsorship (#26) and set it up to provide more time for actual CPR project activities and less for understanding how to manage them.

In addition to initiating new projects during this phase (#23, #24, #27, and #29), the reference group eventually proposed and the steering group approved three core research areas: metrics (#28), modeling (#27), and agile software development (#33). Ericsson divided the funding equally among all three areas to avoid internal competition, with a minor portion reserved for exploring impromptu ideas.

Researchers experienced “waltzing with industry” to be more like a quick-step dance. The top priority one year (for example, #27) wasn't even in the top-10 list the next. Changing market situations, corporate strategies, industry trends, and reorganizations called for frequent course changes. Connecting to a particular Ericsson unit's goals improved important to managing project

tions (#30) helped diffuse project results, and the attention increased pressure on the projects to regularly reflect on, analyze, and present their value to industry. A more mature prioritization of research themes and projects (#31) helped facilitate projects, which increased the planning horizon.

Utilization

Table 4 lists CPR projects and key events during utilization. In this phase, the collaborating partners began directing more attention toward the research's impact on Ericsson's practices and the diffusion of knowledge throughout the company. Because the infrastructure was in place for most of the research activities (#35), the project focus increasingly shifted toward research results.

All ongoing projects (#36, #37, and #39) had found ways to address both needs and goals that had deployment impact while advancing academic knowledge. Project #32's addition of aspects to a large software product's design information, which required model analysis and description that increased understanding of how to improve the product's architecture. The

CPR projects and key events during the formalization phase (2007–2008).

#	Case data	Type	Date	Explanation	Outcome/Effect
23	Project to evaluate adapted SPE, part II	CPR project	Jan. 2007	Methods evaluation research. Restart of #18 with new staffing from ITU.	Project interrupted.
24	Quality-assurance-in-modeling project	CPR project	Jan. 2007	Exploratory empirical research. Goal is to improve Ericsson's quality assurance practices to accommodate model-driven development.	Project concluded. Changed ways of working, one publication.
25	Formalized SAQC setup	Key event	Jan. 2007	SAQC collaboration is formalized with steering group meetings, reference group meetings, progress reports, and budget decisions.	All involved stakeholders know what to do when.
26	Renewed Ericsson sponsorship	Key event	Jan. 2007	Ericsson renews ITU sponsorship and signs a four-year project-based agreement.	Continued CPR.
27	Model-driven architecture (MDA) roadmap project	CPR project	Mar. 2007	Qualitative case study research. Goal is to develop a technology roadmap to guide incremental introduction of MDA at Ericsson.	Project concluded. Influenced strategies; two publications; new CPR project (#32).
28	Metrics project, part II	CPR project	2007	Action research. Goal is to develop measurement system based on ISO/IEC 15939 and introduce it into the organization.	Project concluded. New major practices implemented; one publication; new CPR project (#34).
29	Project in common with EU FP7 grant: "Procedures for Testing"	Key event	Sep. 2007	Granted 2.5M€ EU project, based on #10 and #12. Goal is to study software engineering approaches that use automated testing and trace analysis to improve reliability of service-oriented networks.	Four CPR projects started between several Ericsson locations and ITU.
30	Improved SAQC communication	Key event	Nov. 2007	Communication formalized: project proposals, quarterly status reports, and semiannual poster presentations and public talks.	Increased involvement of stakeholders, engineers, and researchers in SAQC actions and results.
31	Mature prioritization of new research projects	Key event	2008	Stable annual budgets established for three research themes reduced decision effort and increased planning horizon.	Increased steering group involvement in prioritizing what to do.
32	Adding aspects project	CPR project	Apr. 2008	Qualitative case study research. Goal is to improve the understandability and usability of system models by introducing new viewpoints.	Project concluded: Increased understanding of design modeling; one publication; new CPR project (#39).
33	Agile software development project, part I	CPR project	June 2008	Qualitative case study research. Goal is to describe perceptions of streamlined development and the prerequisites that seemed important to act agile.	Project concluded: Increased reflection and theory awareness; one publication; new CPR project (#37).
34	Metrics project, part III	CPR project	2008	Action research. Goal is to implement organizational metrics and prediction systems.	Project concluded: New practices implemented and widely diffused; one publication; new CPR project (#36).

TABLE 4

CPR projects and key events during the utilization phase (2009–present).

#	Case data	Type	Date	Explanation	Outcome/Effect
35	Infrastructure utilization	Key event	2009	Steering group meetings uses for strategic decision making, posters used to communicate, reference group meetings used for discussions and suggestions.	Engaged managers and collaborators to focus on research outcome.
36	Metrics project, part IV	CPR project	2009	Action research. Goal is to establish quantitative early-warning systems in practice.	Project ongoing: New practices for metrics' quality assurance implemented; four publications.
37	Agile software development project, part II	CPR project	2009	Qualitative case study. Goal is to find bottlenecks and lead times that limit customer-value flows, agility, and speed in large teams	Project ongoing.
38	Project in common with Vinnova grant proposal to streamline development infrastructure	Key event	May 2009	Three ongoing SAQC projects align their work into one grant application including cooperation with yet another partner.	Creating bridges between SAQC CPR projects.
39	Viewpoint engineering project	CPR project	Oct. 2009	Action research. Goal is to establish methods for systematically identifying and prioritizing architectural viewpoints.	Project ongoing. Potential influence on architecture model implementation.
40	New SAQC communication poster	Key event	Oct. 2009	SAQC's vision and content is iterated as a result of #35 and aligned to better match Ericsson's current goals.	One coherent way to present SAQC and its ongoing CPR projects.
41	Industry orientation appraisal	Key event	Oct. 2009	International and national awards for SAQC's orientation toward addressing real needs in industry.	Internal (Ericsson) and external (IEEE) recognition of SAQC work.

Likert scale to assess maturity, with 1 representing low maturity, 5 representing high maturity, and 4 representing enough maturity to sustain successful research projects. In cases of dispute, we referred to project data or involved individuals to reach consensus.

Evaluation results show continuous growth in all factors. This was expected, given the increasing CPR knowledge. Results also show that no success factors reached a satisfactory degree of maturity in the initial phase. Moreover, five years passed without satisfying degrees of impact and innovativeness. Despite this, we judged

collaboration during initialization as successful (#26). This indicates that overall success doesn't require all success factors to be met—particularly not for young collaborations.

A CPR Collaboration Model

To understand which success factors were most important to SAQC in each phase, we reflected on the causes and effects of maturation. For each CPR project and key event, we considered which success factors the project or event depended on and the effects they had on the collaboration. Figure 2 visualizes our findings:

- A general and mutual dependency exists between factors related to research results and research activities. The right activities enable the right kind of results, which in turn enable further research activities.
- The leading success factor for research results is “Need orientation,” and the leading success factor for research activities is “Management engagement.”
- A dependency chain exists between results-related success factors, and another exists between activity-related factors. In our case, “Need orientation” enabled “Industry goal

TABLE 5

Success factors for SAQC research activities and results, together with their maturity rated from 1 (low) to 5 (high) for each development phase.

SAQC success factor		Success factor definition	Maturity evaluation by phase			
			Initiation (2002– 2004)	Alignment (2005– 2006)	Formalization (2007– 2008)	Utilization (2009– present)
Research activities	Management engagement	Industry and academia management are engaged in problem formulation and research conduct and management.	3	4	5	5
	Network access	Research project has access to relevant competences and best-in-class employees.	3	4	5	5
	Collaborator match	Project collaborators willingly engage in each other's areas of interest and challenges.	2	3	4	5
	Communication ability	Project communicates ideas, progress, and results.	2	3	4	4
	Continuity	Topics or contexts stay current over longer time periods.	n/a	3	4	4
Research results	Need orientation	Project results address perceived real-life industry problems and possibilities.	3	4	5	5
	Industry goal alignment	Results align to current industry unit goals.	3	3	4	5
	Deployment impact	Results are deployed (as processes, methods, techniques, or knowledge) and have an impact on practice.	2	3	4	4
	Industry benefits	Results give benefits to industry.	2	3	4	4
	Innovativeness	Results generate new ideas, knowledge, patents, processes, methods, techniques, and publications.	2	3	3	4

alignment,” which enabled “Deployment impact,” which enabled “Industry benefits,” which enabled “Innovativeness.” We found a similar chain involving “Management engagement,” “Network access,” “Collaborator match,” “Communication ability,” and “Continuity.”

The maturation model in Figure 2 is not the only possibility. Other situations might involve different enable-

ment logics, such that factors might mature in an order different from SAQC's case. However, a comparison with related research confirms that CPR should address “Need orientation” and “Management engagement” early. In the technology transfer model of Tony Gorschek and his colleagues, step one is to “Identify potential improvements based on industry needs.”¹⁰ In Andrew Van de Ven's engaged scholarship model, step one is to “Situate, ground,

diagnose, and infer the research problem by determining who, what, where, when, why and how the problem exists up close and from afar.”¹¹ Van de Ven recognizes engagement and involvement by both researchers and engineers as key to managing successful collaborations. Tina Barnes and her colleagues⁶ and Eva Mora-Valentin⁷ consider project management central for successful collaborations.

Each of the other success factors

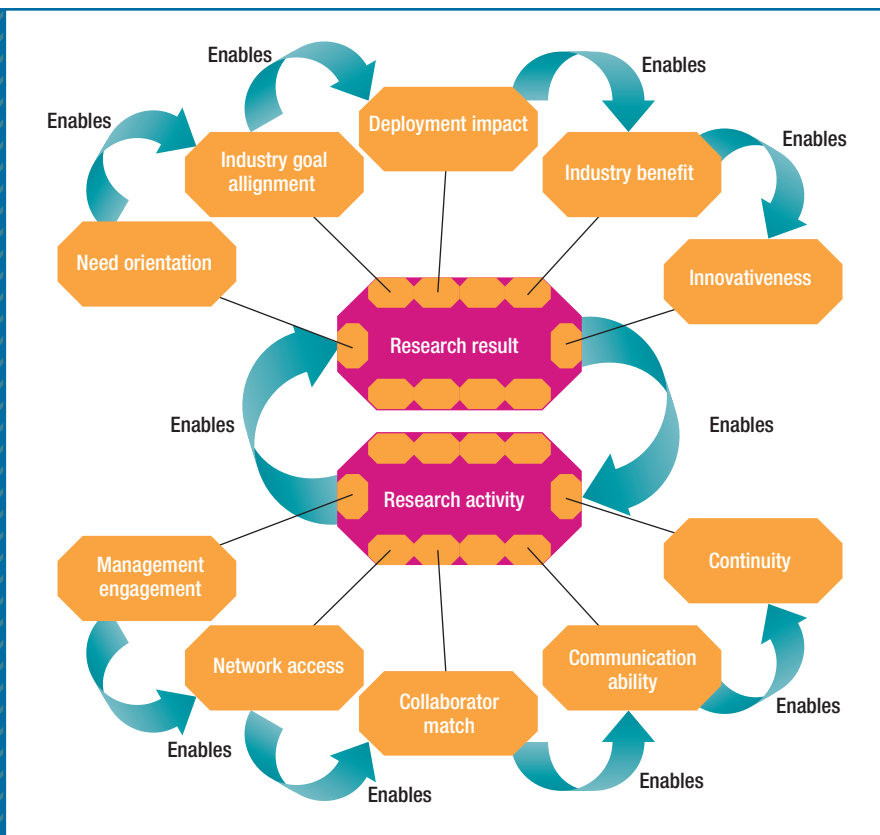


FIGURE 2. A collaboration model for collaborative practice research. Successful research projects must attend to both industry needs and management involvement.

we've identified is supported or partly supported by at least two studies,^{3,7,8,11} but we haven't found recommendations on the order in which to address them.

So, although our model is based on a single (but longitudinal) CPR effort, we have some external support for making recommendations based on our experience.

Making Collaboration Successful

Bringing acting practitioners and reflecting researchers together for CPR establishes opportunities to improve software in ways not possible without it.^{4,11} The collaboration model in Figure 2 lays the foundation for action principles in setting up and managing CPR projects. We've developed 10 action principles to assist anyone involved

or interested in industry-academia collaborations beyond mere technology transfer.

Address Activities to Ensure Results

Industry-academia collaborations ultimately seek industry benefits and innovation. Within the SAQC collaboration, we've seen the value of facilitating activities, such as engaged management and communication, to enable these results.

Ensure Management Engagement

A structured set-up of meetings, participants, reports, budget rules, and so on to engage management is important, as it is in any business. As this engagement matures, managers can concentrate on research activities and project results rather than hunting people down and

discussing when to follow up on what, and so on.

Embrace Research Negotiations

Industry and academia have different objectives. The collaboration must satisfy both parties' needs. Negotiations on directions, goals, and time frames are a natural part of this process. Basing industry funding on small, frequent projects enforces engagement in each others' needs and aligns research with industry goals.

Organize Get-Togethers

Opportunities for skilled engineers and researchers to meet and communicate are crucial for facilitating productive matches between potential collaborators and building an access network to people available for future projects. An organized group should assume responsibility for developing ideas into proposals and deciding which ones to adopt.

Communicate both Progress and Result

Research projects normally don't have immediate deployment impact, so maintaining organizational interest in them requires finding ways to visualize progress as well as results. Communication facilitates continuity, which is one definition of CPR success.

Attend to both Needs and Goals

Collaborations that target an industry need and involve a business unit with an immediate goal related to this need are more likely to succeed than those that address only general needs.

Be Agile

Just as in other industrial projects, CPR projects must deal with an ever-changing business environment. To facilitate results that have deployment impact, projects must accept and respect that industry goals and research directives change. Projects should address only research questions that allow fre-

quent deliveries to industry—about six months seems optimal. Collaborative research must be agile.

Fund Small Research Projects

Industry should distribute funding in small portions to keep research projects well aligned with its needs and goals. This approach supports the frequent course changes needed to be agile.

Allow Innovation to Emerge from Needs

Successful research leads to innovative contributions to knowledge, processes, methods, techniques, and patents. Our experience shows that attention to both deployment impact and industry benefits facilitates innovativeness. Research that reflects on industry practice, working closely with its prioritized needs and goals, is a recipe for success.^{9,12–14}

Realize that CPR Involves Learning

It takes time for both parties to understand each others' needs. Researchers must learn to be agile toward industry needs, and practitioners must learn to appreciate research rigor. Such learning requires time, perseverance, and conscious reflection.

Appreciating each other's strengths is a precondition to getting industry-academia collaborations started. Paying attention to identified success factors improves the collaboration. Accepting that collaborative research must be agile keeps it current and makes collaborations flourish over time. 

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